

ISO/IEC/IEEE 29148:2018 STANDARD

Software Requirements Specification (SRS)

Next-Gen Multi-Tenant Clinic & Hospital SaaS Platform with Subscription Governance & Intelligent Clinical Workflow Engine

PROJECT NAME

MediPortal Cloud (Doctor Prescription & Clinic Suite)

DOCUMENT IDENTIFIER

SRS-MEDIPORTAL-ISO29148-V2.1

ARCHITECTURE FRAMEWORK

Multi-Tenant SaaS (Clinic-Centric Tenant Model)

CORE TECHNOLOGY STACK

PHP 8.3 • Laravel 13.x • MySQL 8 • Bootstrap 5

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Table of Contents

1.0 Executive Summary & Vision	Page 3
2.0 Scope & Compliance Standards (ISO/IEC/IEEE 29148:2018)	Page 3
3.0 Overall Architecture & User Classes	Page 4
4.0 Clinic-Centric Multi-Tenant Subscription Architecture	Page 5
5.0 Subscription Plans & Quota Matrix	Page 6
6.0 Feature Catalog & Modular Access Control	Page 7
7.0 Subscription-Based Login & Session Security	Page 8
8.0 Detailed Functional Requirements by Module	Page 9
8.1 Super Admin & SaaS Governance	Page 9
8.2 Clinic / Hospital Administration Portal	Page 10
8.3 Doctor Portal & Smart Prescription Engine	Page 11
8.4 Receptionist Desk & OPD Queue Management	Page 12
8.5 Staff / Nursing Examination Desk	Page 13
8.6 Patient Revisit & Follow-up Discount Policy	Page 13
9.0 Non-Functional Requirements (ISO/IEC 25010 Quality Model)	Page 14
10.0 Data Model & Entity Relationships	Page 15
11.0 Acceptance Criteria & Verification Protocols	Page 15

1.0 Executive Summary & Vision

MediPortal Cloud is an enterprise-grade, multi-tenant Medical Practice Management and Electronic Prescription SaaS platform. The software empowers outpatient clinics, polyclinics, nursing homes, and multi-specialty hospitals to streamline their clinical operations, digital prescription generation, OPD token queues, vitals tracking, staff administration, and billing.

The platform features an advanced **Clinic-Centric Subscription Governance Engine** that enforces strict plan-based feature provisioning, doctor quotas, staff capacities, and real-time session lockouts. Doctors and clinic staff inherit subscription rights automatically from their parent healthcare facility, providing clean organizational billing and security.

2.0 Scope & Compliance Standards

This Software Requirements Specification (SRS) is structured in compliance with **ISO/IEC/IEEE 29148:2018 (Systems and software engineering — Life cycle processes — Requirements engineering)** and adheres to **IEEE Std 830-1998**.

STANDARD / FRAMEWORK	SCOPE OF COMPLIANCE	IMPLEMENTATION IN SYSTEM
ISO/IEC/IEEE 29148:2018	Requirement structure, completeness, verifiability, and traceability.	Systematic breakdown of functional, non-functional, and interface requirements.
ISO/IEC 27001:2022	Information Security Management System (ISMS) principles.	Role-Based Access Control (RBAC), password hashing via Bcrypt, HTTPS, CSRF shielding.
DISHA & HIPAA Guidelines	Confidentiality and integrity of Electronic Health Records (EHR).	Tenant isolation, strict ownership bounds, audit trails on prescription & correction requests.
ISO/IEC 25010:2011	Software Product Quality & Evaluation Model.	Performance efficiency, sub-second prescription generation, usability, reliability.

3.0 Overall Architecture & User Classes

The application is architected around a **Role-Based Multi-Guard Authentication Matrix**. Every actor operates within a strictly partitioned domain with automated subscription-aware middleware:

ACTOR / USER CLASS	AUTHENTICATION GUARD	PRIMARY RESPONSIBILITIES & CAPABILITIES
Super Administrator	web (role: admin)	Platform governance, SaaS plan creation, feature catalog editing, subscriber management, global metrics, clinic onboarding oversight.
Onboarding Manager	web (role: onboarding)	Clinic credential verification, physical verification checklist, doctor approval, compliance document inspection.
Clinic / Hospital Admin	clinic	Practice operations, medical team onboarding (up to plan quota), staff hiring (up to plan quota), revisit rule configuration, payment categories, prescription branding.
Consulting Doctor	doctor	OPD consultations, digital prescription authoring, formulary lookup, drug interaction alerts, patient history inspection, PDF prescription download.
Receptionist	member (role: receptionist)	Front desk patient registration, token generation, daily OPD queue ordering, revisit policy verification, fee receipt generation.
Clinical Staff / Nurse	member (role: staff)	Patient triage, physical examinations, vitals recording (Blood Pressure, Pulse, Temperature, SpO2, Blood Sugar).

Architectural Principle: Clinic-Held Tenant Model

Individual doctors and staff members **do not purchase or maintain independent subscriptions**. The Clinic/Hospital is the singular billable subscriber entity. When a Clinic's subscription plan is active, all doctors and staff mapped to that clinic inherit access rights seamlessly.

4.0 Clinic-Centric Multi-Tenant Subscription Architecture

The subscription engine enforces monetization, tiering, and operational capacities through four fundamental mechanisms:

- **Automatic Hierarchy Resolution:** The system identifies the calling context (Clinic, Doctor, or Staff) and dynamically traces it back to the parent Clinic instance.
- **Ceiling Quotas (Hard Enforcement):** Prevents resource allocation beyond the purchased tier (e.g. attempting to add a 3rd doctor when the plan allows 2).

- **Feature Flag Provisioning:** Entire functional modules (e.g., Revisit Rules, Staff Accounts) are toggle-gated; unpurchased modules are hidden from UI and blocked at the route level.
- **Real-time Expiration Interceptors:** Time-based validity checks run during login attempts and continuous session requests.

5.0 Subscription Plans & Quota Matrix

The platform comes pre-configured with four structured tiers engineered for different scales of healthcare facilities:

PLAN PARAMETER	CLINIC FREE TRIAL	CLINIC STARTER	CLINIC PROFESSIONAL	CLINIC ENTERPRISE
Target Facility	New Clinics / Pilots	Solo Practices & Small Clinics	Polyclinics & Nursing Homes	Hospitals & Multi-Specialty
Standard Pricing	₹0.00 (14 Days)	₹999.00 / Month	₹1,999.00 / Month	₹3,999.00 / Month
Doctor Capacity	Max 2 Doctors	Max 2 Doctors	Max 6 Doctors	Unlimited (-1)
Staff Capacity	Max 2 Staff	Max 3 Staff	Max 6 Staff	Unlimited (-1)
Monthly Patients	150 Patients/Mo	500 Patients/Mo	Unlimited	Unlimited
Monthly Prescriptions	150 Rx/Mo	500 Rx/Mo	Unlimited	Unlimited
Digital Prescriptions	✓ Included	✓ Included	✓ Included	✓ Included
Patient Directory	✓ Included	✓ Included	✓ Included	✓ Included
Vitals & Physical Exam	✓ Included	✓ Included	✓ Included	✓ Included
Custom Formulary	✓ Included	✓ Included	✓ Included	✓ Included
Payment Categories	✓ Included	✓ Included	✓ Included	✓ Included
Staff & Receptionist Module	✗ Gated	✗ Gated	✓ Included	✓ Included
Patient Revisit Rules	✗ Gated (Hidden)	✗ Gated (Hidden)	✓ Included	✓ Included
Prescription Branding	✗ Gated	✗ Gated	✓ Included	✓ Included
Document Storage Vault	✗ Gated	✓ Included	✓ Included	✓ Included
Analytics & Revenue Reports	✗ Gated	✗ Gated	✓ Included	✓ Included

Client-Specified Gating Rule

If a plan does not include a feature (e.g. `revisit_rules` in Free Trial), the software automatically **hides the section from the navigation sidebar, profile view, and front desk queue**. Attempting to directly type the URL triggers an upgrade interception.

6.0 Feature Catalog & Modular Access Control

The platform defines 10 standardized, discrete feature identifiers that Super Admins can dynamically toggle for any subscription plan:

1. Prescription Writing & PDF Generation

`prescriptions`

Enables doctor consultation desk, smart drug search, dosage calculator, advice snippets, and printable high-definition PDF prescriptions with QR code verification.

2. Patient Management & Medical History

`patients`

Demographic record storage, past visit tracking, symptom archives, and multi-doctor patient history timeline.

3. Vitals & Physical Examination Tracking

`vitals_exam`

Pre-consultation triage module allowing nursing staff to log Blood Pressure, Pulse Rate, Temperature, SpO2, and Random Blood Sugar.

4. Custom Medicine & Formulary Master

`medicines_master`

Custom pharmacology catalog management allowing clinics to create proprietary brand names, generic formulations, default dosages, and intake instructions.

5. Staff & Receptionist Accounts

`staff_management`

Dedicated role-separated logins for front desk receptionists and clinical support staff. Governed by `max_staff` quota ceiling.

6. Consultation Fees & Payment Categories

`payment_categories`

Custom fee category creation (General OPD, Emergency, Specialist Consultation, Procedure Fees) and price configuration.

7. Patient Revisit Rules & Discounts

revisit_rules

Automated policy engine that recognizes returning patients within a clinic-defined window (e.g., 7 days) and automatically waives or discounts consultation fees.

8. Clinic Branding & Prescription Customization

custom_branding

Clinic letterhead upload, custom header/footer layout, doctor digital signature integration, clinic stamp, and custom prescription canvas design.

9. Clinic & Doctor Document Vault

document_storage

Digital repository for clinic registration certificates, medical board licenses, and compliance documents.

10. Practice Performance & Analytics Reports

analytics_reports

Aggregated visual reporting on daily OPD trends, doctor consultation volumes, fee collection totals, and patient retention rates.

7.0 Subscription-Based Login & Session Security

The security boundary guarantees that healthcare facilities cannot access the system after plan expiration. The enforcement operates on two distinct layers:

7.1 Pre-Authentication Gate (Login Controller)

When a user enters credentials on the central /login screen, the system executes the following strict verification sequence:

1. **Credential Validation:** Hash verification via Bcrypt (cost 12).
2. **Account Status:** Checks `isdeleted == 0` and `status == active`.
3. **Verification Check:** Confirms the clinic and doctor/staff accounts are officially verified.
4. **Subscription Interception:**
 - **For Clinic Admins:** Executes `$clinic->currentSubscription()`. If expired, suspended, or missing, login is rejected with: *"Your clinic's subscription expired on DD Mon YYYY. Please renew your plan to log in."*
 - **For Doctors:** Traces the doctor's registered clinic. If the clinic subscription is inactive or expired, login is rejected with: *"Access Denied: Your clinic's ({Clinic Name}) subscription expired on DD Mon YYYY. Doctor login is disabled until renewed."*
 - **For Receptionists & Staff:** Traces parent clinic hierarchy. If the clinic subscription is inactive or expired, login is rejected with: *"Access Denied: Your clinic's ({Clinic Name}) subscription expired on DD Mon YYYY. Staff login is disabled until renewed."*

7.2 Active Session Lockout (Middleware)

For active sessions already open when a subscription reaches its midnight expiration time:

- The `CheckSubscriptionAccess` (alias: `subscription.active`) middleware executes on every protected HTTP request.
- If `SubscriptionService::hasActiveSubscription()` returns false, the middleware immediately invalidates the active session (`Auth::logout()`), clears CSRF tokens, and redirects the user to /login with an explicit expiration notice.

8.0 Detailed Functional Requirements by Module

8.1 Super Admin & SaaS Governance Module

REQ. ID	FEATURE NAME	DETAILED REQUIREMENT SPECIFICATION
REQ-ADM-01	SaaS Plan Builder	Super Admin shall be able to create, edit, activate, and deactivate subscription tiers with custom pricing, billing frequencies (Monthly/Yearly), trial days, and custom feature sets.
REQ-ADM-02	Quota Assignment	Super Admin shall configure numeric quotas for each plan: max_doctors, max_staff, max_patients_per_month, and max_prescriptions_per_month. Value -1 represents unlimited capacity.
REQ-ADM-03	Subscribers Directory	Super Admin shall view all subscribed clinics in a clean tabular view displaying Clinic Name/ID, Current Plan, Validity Period, Days Left, and Real-time Status badges.
REQ-ADM-04	Subscription Extension	Super Admin shall be able to instantly extend any clinic's subscription validity by +30 Days, +90 Days, or +365 Days via a quick dropdown action.
REQ-ADM-05	Status Override	Super Admin shall be able to manually suspend, activate, or terminate any clinic subscription.

8.2 Clinic / Hospital Administration Portal

REQ. ID	FEATURE NAME	DETAILED REQUIREMENT SPECIFICATION
REQ-CLN-01	Doctor Management with Quota Ceiling	Clinic Admin shall register consulting doctors. The system shall evaluate <code>canAddDoctor()</code> against the plan's <code>max_doctors</code> quota. If capacity is full, the Add button is disabled and submission blocked.
REQ-CLN-02	Staff Management with Quota Ceiling	Clinic Admin shall manage Receptionists and Nurses. If plan feature <code>staff_management</code> is disabled, the module is hidden. If feature is enabled, capacity is capped at <code>max_staff</code> .
REQ-CLN-03	Revisit Policy Engine	Clinic Admin shall configure patient revisit rules: Revisit Window (e.g. 7 days) and Fee Type (Free ₹0 or Custom Discount). Gated behind <code>revisit_rules</code> feature.
REQ-CLN-04	Prescription Design & Letterhead	Clinic Admin shall upload logo, digital signature, header/footer assets. Gated behind <code>custom_branding</code> feature.
REQ-CLN-05	Doctor Correction Requests	Clinic Admin shall review, approve, or reject profile correction requests submitted by doctors (qualification updates, license updates).

8.3 Doctor Portal & Smart Prescription Engine

REQ. ID	FEATURE NAME	DETAILED REQUIREMENT SPECIFICATION
REQ-DOC-01	Smart Prescription Generator	Doctor shall author prescriptions with dynamic drug auto-complete, dosage guidelines (e.g. 1-0-1), administration intervals, duration (Days/Weeks), and special advice.
REQ-DOC-02	Chief Complaints & Diagnosis	Doctor shall log patient symptoms, diagnosis tags, and clinical observations from a standardized master catalog.
REQ-DOC-03	Patient History Timeline	Doctor shall view complete chronological visit history, previous prescriptions, and recorded vitals of any patient registered under the clinic.
REQ-DOC-04	High-Definition PDF Export	Doctor shall download printable PDF prescriptions with doctor license, clinic letterhead branding, patient ID, diagnosis, Rx items, and digital signature.

8.4 Receptionist Desk & OPD Queue Management

REQ. ID	FEATURE NAME	DETAILED REQUIREMENT SPECIFICATION
REQ-REC-01	Patient Registration & Lookup	Receptionist shall register new walk-in patients or search existing patients by Phone, Name, or Patient ID.
REQ-REC-02	Daily OPD Queue	Receptionist shall allocate patients to specific consulting doctors, generating today's consultation queue tokens.
REQ-REC-03	Revisit Policy Automation	When registering a returning patient, the system checks if previous visit was within the clinic's revisit window. If valid, it automatically flags the visit as " <i>Free Revisit</i> ".

8.5 Staff / Nursing Examination Desk

REQ. ID	FEATURE NAME	DETAILED REQUIREMENT SPECIFICATION
REQ-STF-01	Triage Vitals Entry	Clinical staff shall record pre-consultation vitals: Blood Pressure (Systolic/Diastolic), Pulse Rate (bpm), Body Temperature (°F), SpO2 (%), and Blood Sugar (mg/dL).
REQ-STF-02	Doctor Sync	Recorded vitals shall reflect instantaneously in the doctor's consultation dashboard for that patient's visit.

9.0 Non-Functional Requirements (ISO/IEC 25010 Quality Model)

QUALITY CHARACTERISTIC	MEASURABLE METRIC & ARCHITECTURAL IMPLEMENTATION
Performance Efficiency	• Prescription generation rendering time shall be < 500 milliseconds. • Database queries optimized with compound indices on <code>clinic_id</code>, <code>doctor_id</code>, and <code>patient_id</code>. • High-throughput PDF rendering via optimized headless engines.
Security & Confidentiality	• Bcrypt password hashing algorithm with individual salting. • Cross-Site Request Forgery (CSRF) protection on 100% of modifying HTTP requests. • SQL injection immunity guaranteed via Eloquent ORM parameterized queries. • Cross-Site Scripting (XSS) auto-escaping across all Blade template variables.
Reliability & Availability	• 99.9% target operational availability. • Session invalidation and graceful fallback on subscription expiry without data loss. • Soft delete protection (<code>isdeleted</code>) preventing accidental permanent record loss.
Usability & UX	• Clean, high-contrast Valex medical dashboard aesthetic. • Color-coded KPI counters for immediate operational visibility. • Mobile, tablet, and widescreen responsive layouts.

10.0 Entity Relationship Model Overview

The data model is structured around a central `clinics` table holding 1-to-N relationships with `doctors`, `members (staff)`, `patients`, `subscriptions`, and `payment_categories`. Subscriptions bind to `subscription_plans`, which dynamically aggregate records from the `subscription_features` master catalog.

11.0 Acceptance Criteria & Verification Sign-off

TEST CASE ID	VALIDATION PROTOCOL	EXPECTED OUTCOME	STATUS
TC-SUB-01	Clinic with active subscription attempts login.	Login succeeds; redirects to clinic.dashboard.	Passed
TC-SUB-02	Clinic with expired subscription attempts login.	Login rejected; error shows exact expiry date with renewal advice.	Passed
TC-SUB-03	Doctor belonging to active clinic attempts login.	Login succeeds; inherits active clinic plan.	Passed
TC-SUB-04	Doctor belonging to expired clinic attempts login.	Login blocked with notice that clinic subscription has expired.	Passed
TC-SUB-05	Clinic reaches max_doctors quota limit.	Add Doctor button locks; controller rejects creation with quota alert.	Passed
TC-SUB-06	Clinic plan lacks revisit_rules feature.	Revisit menu completely hidden; route access safely intercepted.	Passed

Document Sign-off & Delivery Statement

This document represents the official Software Requirements Specification (SRS) for the MediPortal Cloud SaaS Platform. It provides clients, stakeholders, and implementation teams with complete architectural clarity, subscription policies, and functional specifications under international standards.